

Project 02 — AI Music & Audio Analyzer

Updated Test & Development Report — v1.2

Item	Current Value
Project Status	Steps 1–20 validated
Documentation Version	v1.2
Platform	Windows Laptop
Python	3.14.7
Hardware	Windows Laptop + Airdopes Joy Bluetooth Earbuds
Music Input	test_music.mp3
DSP Milestone	Steps 1–17 complete
AI Milestone	Steps 18A–19 complete
Final Reporting	Step 20 complete
Audio Health Score	96 / 100 — Excellent
Final Report	final_ai_music_analysis.html + .json

1. v1.2 Update Summary

This update records the successful completion of Step 20 — Final AI Music Analysis Report Generator. The project now produces a consolidated final HTML report and JSON data package containing the DSP measurements, music analysis, Audio Health Score, Gemini AI interpretation, AI recommendations and Step 18C dashboard reference.

2. Step 20 — Final Reporting Layer

Step 20 is the final aggregation layer of the current Project-02 workflow. It consumes the validated outputs from Steps 13–19 instead of recalculating the audio. This keeps the final report reproducible and separates measurement, AI interpretation, recommendations and presentation.

Input / Output	Purpose
music_feature_summary.json	Measured DSP and music feature package
music_ai_interpretation.json	Structured Gemini interpretation
music_recommendations.json	AI insights and recommendations
ai_music_analysis_dashboard.png	Integrated visual dashboard from Step 18C
final_ai_music_analysis.html	Final human-readable analysis report
final_ai_music_analysis.json	Final consolidated machine-readable package

3. End-to-End Final Architecture

Airdopes / Music File → Audio Loading → DSP Analysis → Music Analysis → Audio Health Score → AI Feature Package → Gemini Interpretation → AI Recommendations → AI Dashboard → Final HTML + JSON Report

4. Final Verified Track Results

Parameter	Result
Track	test_music.mp3
Duration	240.02 sec
Sample Rate	44,100 Hz
Channels	1 (mono)
Estimated Key	G Minor
Key Correlation	0.7472
Estimated BPM	109.96
Tempo Category	Moderate
Dominant Band	Bass
Bass / Mid / Treble	89.42% / 10.50% / 0.08%
RMS Level	-17.05 dBFS
Peak Level	-0.17 dBFS
Dynamic Range	16.88 dB
Clipping	0 samples
Health Score	96/100 — Excellent

5. AI Interpretation & Recommendation Layer

Step 18B interprets the measured features while Step 19 converts those measurements and interpretations into actionable insights. The AI correctly identified the strong bass concentration, limited peak headroom, mono channel configuration, zero detected clipping and the need to manually verify the estimated key and BPM.

AI Area	Verified Result
Overall Insight	Technically clean signal with strong low-frequency concentration
Strength	0 clipped samples
Strength	16.88 dB measured dynamic range
Strength	0.7472 correlation for estimated G Minor
Area to Review	89.42% bass energy / 0.08% treble energy
Area to Review	-0.17 dBFS peak headroom
Area to Review	Mono channel configuration
Priority	Review frequency balance
Priority	Perform true-peak evaluation before final encoding
Priority	Manually verify key and BPM

6. Final HTML Report

The Step 20 HTML report consolidates the project into a browser-readable technical report. It contains track overview, health score, signal analysis, frequency and spectral analysis, rhythm, musical key, the Step 18C dashboard, AI interpretation, AI recommendations, engineering limitations and final conclusion.

7. Final JSON Package

The final JSON package provides a machine-readable representation of the consolidated analysis. It includes track information, signal analysis, spectral analysis, frequency balance, rhythm, musical key, clipping, audio quality, AI interpretation, AI recommendations and references to the final dashboard/report.

8. Completed Project Steps

Step	Feature	Status
1	Audio Device Detection	PASS
2	Airdopes Microphone Verification	PASS
3	Airdopes WAV Recording	PASS
4	WAV Audio Analysis	PASS
5	Waveform Visualization	PASS
6	Audio Level Analysis	PASS
7	FFT / Frequency Spectrum	PASS
8	STFT / Spectrogram	PASS
9	Spectral Feature Analysis	PASS
10	Bass / Mid / Treble Energy	PASS
11	Tempo / Beat Analysis	PASS
12	Integrated Audio Feature Summary	PASS
13	Real Music File Input & Analysis	PASS
14	Professional Music Visualization Dashboard	PASS
15	Key / Chroma Analysis	PASS
16	Loudness / Dynamic Range / Clipping	PASS
17	Audio Quality & Health Score	PASS
18A	Music Feature Package	PASS
18B	AI Music Interpretation	PASS
18C	AI Report & Dashboard Integration	PASS
19	AI Music Recommendation & Insight Engine	PASS
20	Final AI Music Analysis Report Generator	PASS

9. Current Project Structure

```
Project-02-AI-Music-Audio-Analyzer/  
■■■ audio/ → airdopes_test.wav  
■■■ music/ → test_music.mp3  
■■■ reports/ → DSP graphs, dashboards, AI outputs and final reports  
■■■ src/ → Steps 1–20 Python scripts  
■■■ tests/  
■■■ venv/  
■■■ requirements.txt  
■■■ .gitignore  
■■■ README.md
```

10. Security & Reproducibility

The Gemini API key is supplied through the GEMINI_API_KEY environment variable and is not embedded in the project source. Secrets and local environment files should remain excluded from GitHub. Step 20 consumes previously generated

JSON packages, allowing the final report to be reproduced without rerunning Gemini unnecessarily.

11. Engineering Limitations

DSP measurements depend on selected algorithms, sample rate, mono conversion and frequency-band definitions. Key detection is an estimate based on chroma/profile correlation. The Audio Health Score is a project-specific heuristic and is not an industry-standard audio quality metric. AI interpretations and recommendations should be treated as analysis assistance and verified through appropriate listening and engineering measurements.

12. Final Project Conclusion

Project-02 has successfully progressed from Bluetooth audio-device validation and Airdopes microphone recording to a complete AI-assisted music analysis workflow. The system now performs DSP analysis, music feature extraction, musical-key and rhythm estimation, loudness and clipping analysis, health scoring, Gemini interpretation, AI recommendations, dashboard visualization and final consolidated reporting.

The verified test track achieved a project-specific Audio Health Score of 96/100 (Excellent), with zero detected clipping, 109.96 BPM estimated tempo, G Minor estimated key and a strongly bass-dominant frequency distribution. The final reporting layer provides both human-readable HTML and machine-readable JSON outputs.

13. Version History

- v0.1 — Project-02 documentation baseline created from the beginning.
- v0.2 — Steps 4–6: WAV analysis, waveform visualization and audio-level analysis.
- v0.3 — Steps 7–8: FFT spectrum and STFT spectrogram.
- v0.4 — Step 9: spectral feature analysis.
- v0.5 — Steps 10–11: Bass/Mid/Treble and Tempo/Beat analysis.
- v0.6 — Steps 12–13: integrated feature summary and real music track analysis.
- v0.7 — Step 14: professional music visualization dashboard.
- v0.8 — Steps 15–16: key/chroma plus loudness, dynamic range and clipping analysis.
- v0.9 — Step 17: Audio Quality & Health Score; 96/100 — Excellent.
- v1.0 — Steps 18A–18B: AI-ready feature package and Gemini AI Music Interpretation.
- v1.1 — Step 18C and Step 19: AI dashboard integration plus AI recommendation/insight engine.
- v1.2 — Step 20: Final AI Music Analysis Report Generator successfully completed.

14. Current Milestone

Milestone	Status
Steps 1–17 — DSP & Music Analysis	COMPLETE
Steps 18A–18C — AI Feature, Interpretation & Dashboard	COMPLETE
Step 19 — AI Recommendations	COMPLETE
Step 20 — Final Report Generator	COMPLETE
Project-02 Current Status	FINAL REPORTING MILESTONE COMPLETE